

STATS 503, Homework 7

1. [6 POINTS] ISLP chapter 8 exercise #5
2. [6 POINTS] Consider using gradient boosting to solve a regression problem. Suppose that at each iteration, we fit the residuals using a “decision stump,” that is, a decision tree with exactly two leaf nodes. Then, the final estimate of the regression function can be expressed as

$$\hat{f}(x) = \sum_{j=1}^p \hat{f}_j(x_j),$$

where, for $j = 1, \dots, p$, f_j is an arbitrary function of predictor j . Explain why this is so.

3. [6 POINTS] Create a sketch that illustrates the connection between a decision tree and the partition of input space that is implied by that decision tree. You can do this on paper (and scan it) or use a drawing program.
 - (a) Draw an example partition of a two-dimensional feature space that could result from a tree with binary splits, with each feature defined on the interval $[-1, 1]$. Include at least 5 regions.
 - (b) Draw a decision tree corresponding to the partition you drew, labeling every tree branch. Show terminal nodes as rectangles and interior nodes as circles. Do not include predictions at the terminal nodes, just the branch labels (e.g. $X_1 < 0.5$).
 - (c) Draw the path of a particular point, with $X_1 = X_2 = 0.5$, through your tree and label the terminal node where it ends up. Mark this point in your plot from part (a) as well.
4. Consider the `crabs` dataset, which is available from Canvas. This dataset contain five size-related measurements of two different species of crabs, blue and orange. 50 male and 50 female crabs of each species are measured. We will classify species based on the other predictor variables and evaluate errors in terms of the misclassification rate.
 - (a) Use `sklearn.model_selection.train_test_split` to split the data. Set the random state to 6789, the size of the training set to 80% of the size of the entire dataset, and stratify according to species and sex so that approximately the same proportions of species/sex combinations appear in both the held-out and the training dataset.
 - (b) [3 POINTS] Train a classification tree to predict `species` from the five numerical measurements and sex. Select the `max_leaf_nodes` by cross-validation, considering the values $\{2, 3, \dots, 10\}$. Plot the tree. Comment on which variables are used by the tree. Compute training and test errors.

Remark 1: In this case, the use of more than 10 leaf nodes actually leads to better predictive performance, but a single decision tree is more interpretable.

Remark 2: A tree is less interpretable if it has many leaves, so for this problem, please restrict your attention to smaller trees.

- (c) **[3 POINTS]** Now, train a random forest with these data, using $m = 5$ randomly selected predictors for each split and 1000 trees in total. Make a variable importance plot and compare with your results for a single tree. Compute training and test errors.
- (d) **[4 POINTS]** Fit `sklearn.ensemble.HistGradientBoostingClassifier` to the data. Plot the training and test error as a function of the number of trees M constructed by boosting, for a range of values of M up to 1000. *Hint: Fit the tree only once using `max_iter=1000` and then use `staged_predict` to calculate the error at each iteration.* Report training and test errors for a value of M of your choice and justify why you chose that value.
- (e) **[2 POINTS]** Comment on which method appears to perform best for this dataset and whether the results (that is, training and test error) are consistent among the methods.

Total points for this assignment: 30